$$= \frac{x^2 - 11x + 18}{(x-1)(x-2)(x-3)(x-5)} = \frac{(x-9)(x-2)}{(x-1)(x-2)(x-3)(x-5)}$$

$$= \frac{x-9}{(x-1)(x-3)(x-5)}$$



Exercise 3.6

1. Simplify (i) $\frac{x(x+1)}{x-2} + \frac{x(1-x)}{x-2}$ (ii) $\frac{x+2}{x+3} + \frac{x-1}{x-2}$ (iii) $\frac{x^3}{x-y} + \frac{y^3}{y-x}$
2. Simplify (i) $\frac{(2x+1)(x-2)}{x-4} - \frac{(2x^2-5x+2)}{x-4}$ (ii) $\frac{4x}{x^2-1} - \frac{x+1}{x-1}$
3. Subtract $\frac{1}{x^2+2}$ from $\frac{2x^3+x^2+3}{(x^2+2)^2}$
4. Which rational expression should be subtracted from $\frac{x^2+6x+8}{x^3+8}$ to get $\frac{3}{x^2-2x+4}$
5. If $A = \frac{2x+1}{2x-1}$, $B = \frac{2x-1}{2x+1}$ find $\frac{1}{A-B} - \frac{2B}{A^2-B^2}$
6. If $A = \frac{x}{x+1}$, $B = \frac{1}{x+1}$, prove that $\frac{(A+B)^2 + (A-B)^2}{A \div B} = \frac{2(x^2+1)}{x(x+1)^2}$
7. Pari needs 4 hours to complete a work. His friend Yuvan needs 6 hours to complete the same work. How long will it take to complete if they work together?
8. Iniya bought 50 kg of fruits consisting of apples and bananas. She paid twice as much per kg for the apple as she did for the banana. If Iniya bought ₹ 1800 worth of apples and ₹ 600 worth bananas, then how many kgs of each fruit did she buy?

3.5 Square Root of Polynomials

The square root of a given positive real number is another number which when multiplied with itself is the given number.

Similarly, the square root of a given expression $p(x)$ is another expression $q(x)$ which when multiplied by itself gives $p(x)$, that is, $q(x)$. $q(x) = p(x)$

So, $|q(x)| = \sqrt{p(x)}$ where $|q(x)|$ is the absolute value of $q(x)$.

The following two methods are used to find the square root of a given expression

(i) Factorization method (ii) Division method

Progress Check

1. Is $x^2 + 4x + 4$ a perfect square?
2. What is the value of x in $3\sqrt{x} = 9$?
3. The square root of $361x^4y^2$ is _____.
4. $\sqrt{a^2x^2 + 2abx + b^2}$ = _____.
5. If a polynomial is a perfect square then, its factors will be repeated _____ number of times (odd / even).

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